Understanding Diabetes Progression

Data Analytics and Statistics – Final Project, Scenario 1

This report examines which patient characteristics are associated with diabetes progression, what the regression models show, and what conclusions can reasonably be drawn from the data. The analysis was carried out in R.

1. Introduction and data

The dataset contains 442 patients and 11 variables, with one row for each patient. The outcome is progression (originally named Y), a measure of how much the disease progressed one year after the first measurement: higher values mean greater progression. The ten predictors were measured at the start: age, sex, BMI, average blood pressure (bp) and six blood measurements (s1 to s6).

All variables are numerical except sex: the codes 1 and 2 represent categories, so sex was converted into a categorical variable (factor). The two groups are balanced (235 and 207 patients). There are no missing values, no duplicated rows and no unrealistic values. Two minor limitations are that it is not known which sex each code represents, and that the blood measurements only have coded names.

Health data are sensitive and protected by law (for example, the GDPR). Even without names, a combination of variables could help identify individuals. The results must also be interpreted with care, because an association does not prove causation.

2. Exploratory analysis

Progression has a mean of 152 and a median of 140.5, with a large spread (standard deviation 77, range 25 to 346, IQR 124.5). Most patients have values around 60 to 100, but the distribution is wide and slightly right-skewed (Figure 1), and the boxplot shows no outliers (Figure 3).

BMI has a mean of 26.4 and a median of 25.7 (range 18 to 42.2), so more than half of the patients are overweight. Its distribution is slightly right skewed, with a few very high values (above 39) that are realistic and were kept (Figure 2). Age (mean 48.5, range 19 to 79) is roughly symmetric.

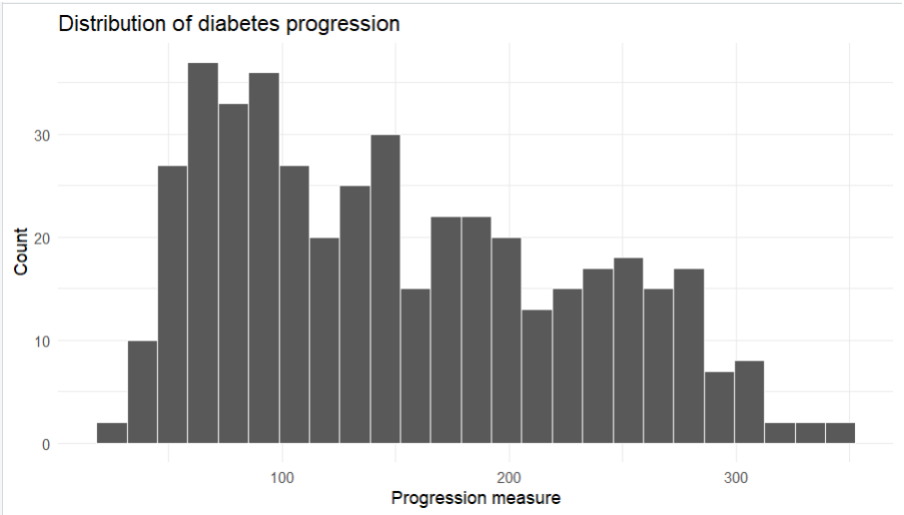


Figure 1: *Distribution of diabetes progression*

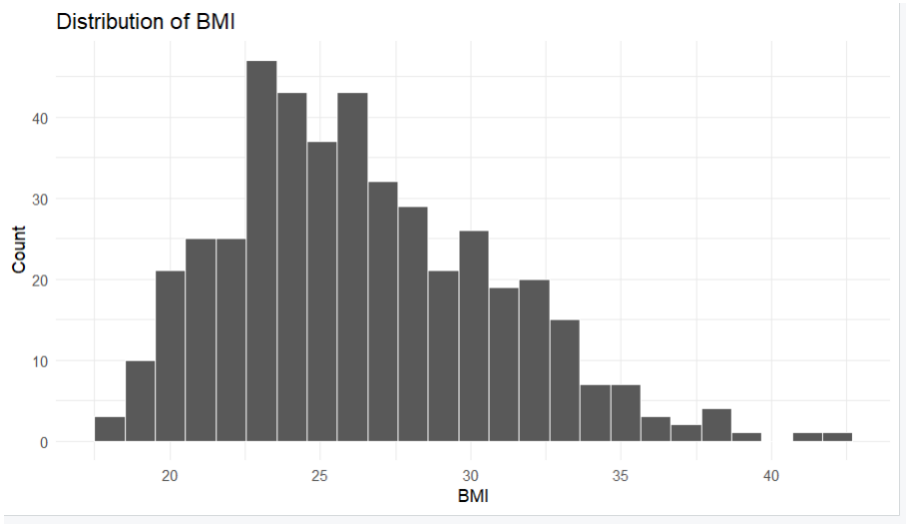


Figure 2: *Distribution of BMI*

Before modelling, it is important to note that the outcome varies a lot between patients and that the variables are measured on very different scales, so the regression coefficients cannot be compared directly with each other.

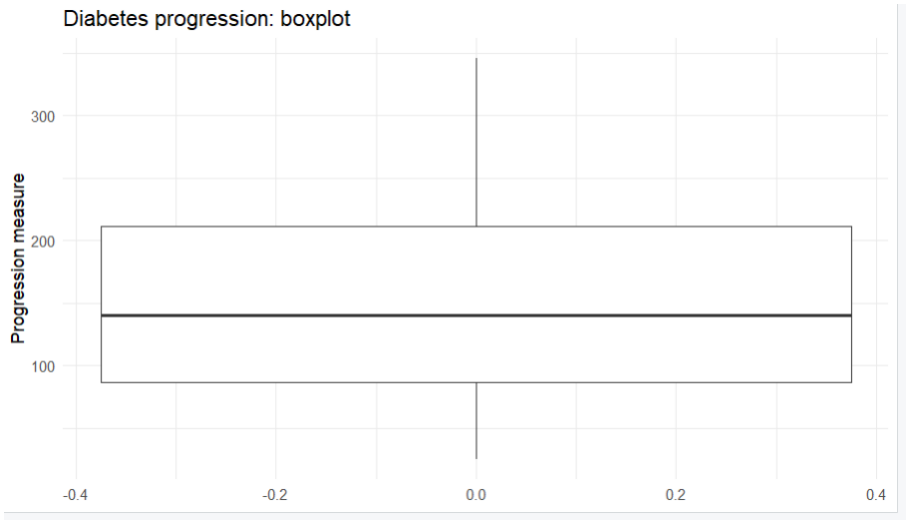


Figure 3: *Boxplot of diabetes progression*

3. Relationships between variables

The correlation matrix (Figure 4) shows that progression is most strongly related to BMI (r = 0.59) and s5 (r = 0.57), and moderately related to blood pressure (r = 0.44) and s4 (r = 0.43). s3 is the only negative relationship (r = −0.39), while age, s1 and s2 are only weakly related (r = 0.17 to 0.21). Sex was not included, because it is a categorical variable. The scatterplots (Figures 5 and 6) show a positive, roughly linear relationship with BMI and blood pressure, but the points are widely scattered around the line.

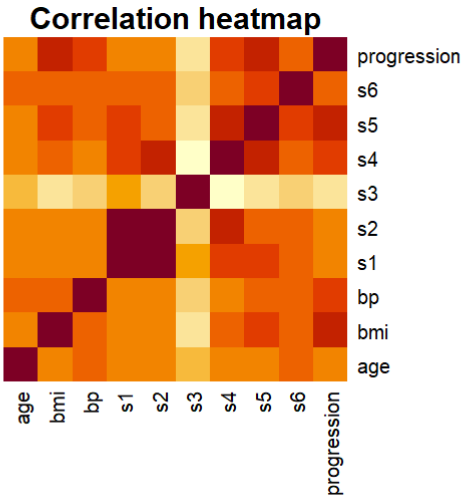


Figure 4: *Correlation heatmap*

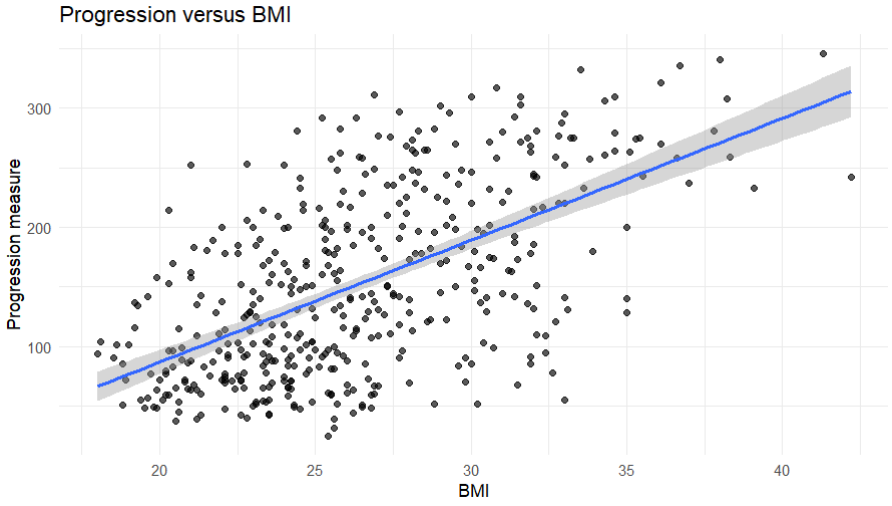


Figure 5: *Progression versus BMI*

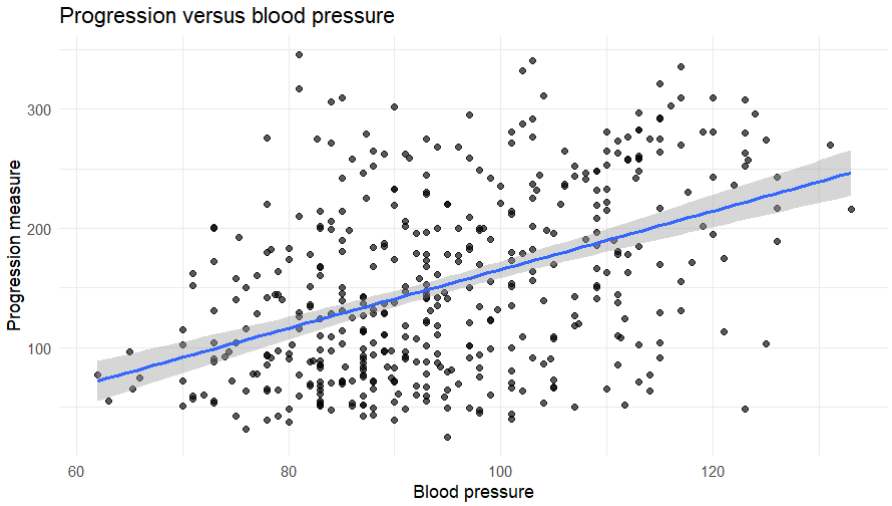


Figure 6: *Progression versus blood pressure*

Some blood measurements are strongly related to each other, mainly s1 with s2 (r = 0.90) and s3 with s4 (r = −0.74). They partly contain the same information and, if used together in a model, could cause multicollinearity. The most promising variables are BMI, s5 and blood pressure, because they are strongly related to the outcome but only moderately related to each other (r = 0.40 to 0.45).

4. Regression modelling

The simple model uses BMI, which had the strongest correlation: progression = −117.77 + 10.23 × BMI. Each additional unit of BMI is associated with about 10 units more progression on average (95% CI: 8.9 to 11.6, $p < 0.001$). The intercept has no practical meaning, because it corresponds to a BMI of 0. R^2 = 0.34 means that BMI explains about 34% of the variation in progression.

The multiple model includes BMI, blood pressure and s5, and all three are statistically significant (Table 1). Holding the other predictors constant, one unit of BMI is associated with about 6.5 units more progression, and 10 units of blood pressure with about 9 units more. The BMI coefficient is smaller than in the simple model, because part of its relationship with progression is shared with blood pressure and s5. The large coefficient of s5 (49.6) is due to its small scale (3.3 to 6.1). For an increase of one standard deviation, BMI (+29) and s5 (+26) have a similar effect, while blood pressure has about half (+12).

Table 1: *Coefficients of the multiple regression model*

Predictor	Estimate	95% CI	p-value
BMI	6.50	5.13 to 7.87	< 0.001
Blood pressure	0.90	0.48 to 1.33	< 0.001
s5	49.58	38.0 to 61.2	< 0.001

The multiple model explains more of the variation (R^2 0.48 vs 0.34, adjusted R^2 0.477 vs 0.342) and reduces the typical prediction error from 62.5 to 55.8. However, the improvement is moderate: about half of the variation remains unexplained, and the model is more complex, with s5 harder to interpret in practice than BMI.

5. Model evaluation and uncertainty

In the residuals plot (Figure 7), the residuals are generally spread randomly around zero, with no extreme values. There is only a mild curve at low predicted values, where the model slightly underestimates progression, but no serious violation of the model assumptions. The VIF values are very low (1.27 to 1.35, well below the common limit of 5), which agrees with the moderate correlations in Section 3. Therefore, multicollinearity is not a problem.

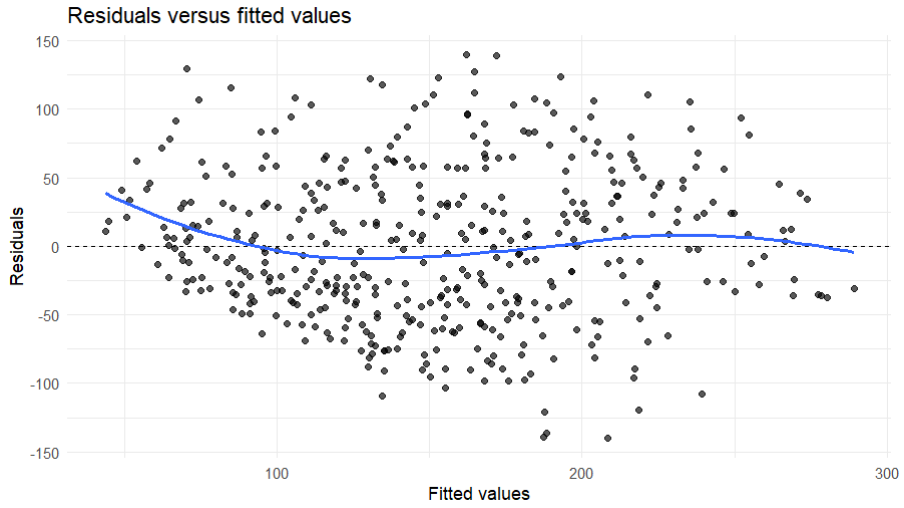


Figure 7: *Residuals versus fitted values*

The BMI coefficient is estimated with good precision (6.50, 95% CI: 5.13 to 7.87). Blood pressure is also significant (0.90, 95% CI: 0.48 to 1.33), but its interval is wider relative to the estimate, so its true effect could be anywhere from small to moderate.

Statistical significance does not necessarily mean practical importance: with 442 patients, even small effects can be significant. For example, 10 units of blood pressure correspond to only about 9 units of progression (the standard deviation of the outcome is 77), while a change in BMI from 25 to 30 corresponds to about 32 units. With a typical error of about 56 units, the model is not suitable for predictions for individual patients.

The main weaknesses are that about half of the variation remains unexplained, that there is mild non-linearity, that the predictors were chosen using the same data, and that the model was not tested on new data. We can therefore trust the direction and approximate size of the relationships, but not individual predictions.

6. Causality, limitations and conclusions

The data are observational: the researchers recorded the patients' characteristics without controlling them, as would happen in an experiment. Therefore, the coefficients show associations, not causes. Patients with higher BMI had greater progression, but this does not prove that BMI causes it.

Important confounding variables are missing from the data, such as medication, diet, physical activity, how long the patient has had diabetes, and smoking. There are also measurement issues: blood pressure was measured only once, BMI is an imperfect measure of body fat, and the way the outcome was calculated, and the coding of sex are not documented. In addition, the results come from 442 patients in a single study and may not apply to other countries, to patients with other treatments, or outside the range of values observed.

The data supports that, in this sample, higher BMI, blood pressure and s5 are associated with greater disease progression, and that together they explain about half of the variation. They do not support claims that reducing BMI will slow the disease by a specific amount, that the model can reliably predict progression for individual patients, that the variables not included are irrelevant, or that the findings apply to all patients with diabetes.

In summary, BMI and s5 are the characteristics strongly associated with diabetes progression, while blood pressure has a smaller but significant relationship. The findings may help identify patients who need closer monitoring, but causal conclusions would require experimental data or data that include the main confounding variables.